Avian Nutrition (VET437)

Welcome to the avian nutrition portion of VET437! This lecture aims to provide you with a solid foundation on the importance of balanced nutrition in companion avian species. Though hundreds of bird species each have unique requirements, we hope to provide you with a clear understanding of the needs of commonly seen species housed in captive settings. This will help you guide and advise owners on their birds' nutritional needs and hopefully avoid more frequently encountered nutrition-related illnesses.

Nutrition Basics:

Birds, like mammals, have basic requirements for macronutrients (water, protein, fats, carbohydrates) and micronutrients (vitamins and minerals). These building blocks are used to make energy to drive biochemical processes. Specific requirements for each group will depend on physiologic and medical status (including species and variations in body size, age, gender, season, and systemic health). Each nutrient has a specific function and is necessary. However, the key to a healthy diet is to provide balance rather than excess or deficiency. Each bird species has adapted to its specific habitat using a different diet strategy. As such, nutritional requirements will vary based on a diet strategy, and one diet will NOT fit every bird. It is essential to think about where your patients are coming from and what foods are native to them when you are making assessments and recommendations. Fruits and vegetables native to North America may not have the same nutrient content or mineral balance they would be getting in the wild. Even the types of seeds we offer in captive diets have very different nutrient compositions than those a bird might typically find in the wild.

Most of our psittacine (parrot) species are considered florivores, meaning they consume plant-based diets and NOT just seeds. This group can then be subclassified into a few more specialty strategies:

- Granivores = Consume seeds/beans, grains, and hard dry fruit like nuts
- Frugivores = Consume primarily fruits (fleshy fruits, not legumes)
- Nectarivores = Nectar eaters
- Omnivores = Consume both animal and plant proteins

Other dietary strategies in avian groups include:

- Piscivores = Consume fish-based diets
- Carnivores = Consume animal proteins primarily

Current diet recommendations and formulated diets are based on standard nutritional requirements for poultry, as this is the area where we have the most scientific information. Information on poultry has been supplemented by work on captive budgerigars, cockatiels, and Amazons. Obtaining accurate information on diet consumption in the wild is challenging. Even within one group, varying dietary strategies will change nutritional requirements, and we don't have enough research to know what is genuinely required for each species, including requirements at various life stages. In the meantime, we make assumptions and extrapolations from the few studies that have been done. Compared to poultry, psittacines have lower calcium, protein, and fat requirements, as they are non-production birds. However, their vitamin and mineral requirements are very similar. These requirements will change based on physiologic state, with marked variations during more demanding scenarios such as molting, growth, reproduction, and illness.

Guidance for Feeding Companion Birds:

A thorough history is crucial when evaluating a sick bird or providing assessments and recommendations during a wellness examination. Some questions to ask include:

1. What is the owner offering? This should include the types of food (seed mix, pellets, fruits, veggies, nuts), the brand, and the amount, proportion, and frequency of each offered.
2. Are supplements provided? If yes, what kind and how are they offered? An important aspect to be aware of is that supplements provided in water will have different consumption and efficacy than those sprinkled over food or provided in tablet form.
3. What is actually consumed by the bird? **This is the most critical question** because owners may offer an appropriate diet, but birds will select the foods they like (much like us!). This often involves selecting foods with higher calories or fat (such as nuts or sunflower seeds). Self-selection can result in an overall inappropriate diet and nutritional imbalances.

4. How is food presented? It is very common for owners to offer large quantities of various food items to allow behavioral selection during the day. But this can also result in imbalances. Owners may also hide food items to provide foraging and enrichment throughout the day or feed only twice per day with no free feeding allowed in between. Many birds are neophobic (fear of something new), so adding new foods or transitioning to more appropriate diets can be challenging if done too rapidly. It is crucial to provide varieties of foods, sizes, textures, and colors early in life to help prevent neophobia from occurring.
5. Does the bird have a favorite food? This can often be utilized when introducing foraging experiences or training. It can also help encourage birds to eat in a novel environment, like the hospital.
6. Diet and parents' preferences may also affect diet choices in young birds that are not hand-reared.

Species-Specific Recommendations:

Psittacines

The most common captive birds seen in a clinical setting are psittacines (parrot species). Specific requirements for this group pertain primarily to water, protein, lipid, and calcium contents. Total water requirements are higher in younger birds (especially at hatch), larger birds, and species from warmer environments. Temperatures above thermoneutral zones for any species will increase water requirements. Protein requirements for maintenance in adult birds positively correlate with body size, so larger birds have higher protein requirements. Protein deficiency is more of a concern than excess unless renal disease/dysfunction is already present. High protein diets have NOT been shown to cause renal disease. However, rapid changes from diets low in protein to those high in protein may contribute. Approximately 4-5% is the dietary lipid amount required to allow the absorption of fat-soluble vitamins and to act as an immediate energy source. Straying above 5% could lead to the formation of issues such as overweight or obese body conditions. Regarding vitamins and minerals, calcium requirements will increase with growth and reproduction but can also vary by species. African grey parrots, for example, tend to have higher dietary calcium needs than other psittacine species of similar size. Vitamins and other minerals often need to be supplied in the diet, but the specific amount varies by nutrient.

Passerines

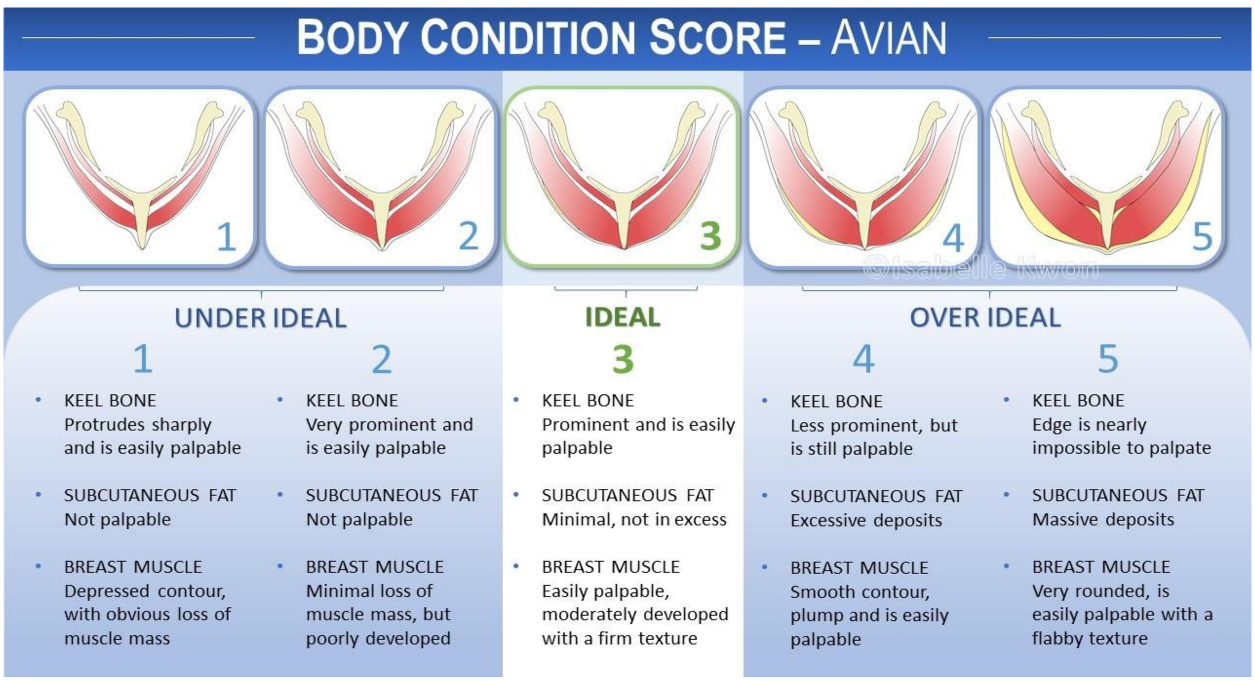
The passerines include small birds like canaries and finches, as well as Mynah birds, and generally have higher energy requirements than psittacine birds. Passerines are typically granivores and can handle a 100% seed diet much better than psittacines (even though this is still not recommended). Again, available seed mixes still do not contain the same seeds these birds would find in the wild, thus not optimal. The recommended diet is about 50% seed and commercial pelleted diet mix to provide the proper vitamin and mineral balance and 50% fresh produce. They can also occasionally be offered insects as treats. It has been noted that passerines process carotenoids better in the diet than non-passerines. Carotenoids contribute to red, orange, and yellow feather coloration and can be provided by utilizing produce high in these substances (carrot juice, orange-colored produce, etc.). Breeders often use high carotenoid diets to have more brightly colored birds for expositions.

Columbiformes

The order of Columbiformes includes pigeons and doves, often kept in a captive setting as companion animals and as racing birds. These species are generally considered granivores, although some more frugivorous species are also present. In the wild, granivorous species feed on various seeds, legumes, fruits and berries, young leaves and buds, flowers, and some invertebrates (insects, snails, earthworms). They generally swallow seeds whole as they forage on the ground, so they play an important role in dispersing seeds. Columbiformes drink water differently than most birds. They drink actively by suction by placing their beak in water up to their nares. Most other birds take in water and throw their head back to swallow it. Granivores should be given a mix of a commercially available pellet or crumble, seeds, and legumes. Diets are available that are specifically formulated for pigeons. Small amounts of vegetables and insects can also be provided. Columbiformes require grit as they swallow seeds whole and need help grinding them up. Vitamin and mineral supplements, especially calcium, should be considered for laying females and racing birds with higher energy demands.

Assessing Nutritional Status:

Physical examination can provide a lot of information on the current nutritional status of our avian patients in combination with information from their diet history and overall husbandry. Weight and body condition scores can help determine malnutrition/loss of condition and over-conditioning/obesity. Various body condition scoring systems are used, but the most important part is maintaining consistency in evaluating our patients. It is important to consider pectoral muscle mass, how the keel palpates, and fat deposits along the body (particularly along the coelom, proximal limbs, and scapular areas). Assessing overall feather quality and color and evaluating for stress bars is another important part of the exam. As discussed previously, feather production requires a lot of energy and protein, and deficiencies can lead to dull coloration, easily broken feathers, or bars reflecting abnormalities in development. These birds may also exhibit signs of feather destructive behavior or abnormal feather growth. Obtaining photos on initial examination can help better monitor for changes as dietary adjustments are made. Traditionally, avian body condition is scored on a scale of 1-5. However, we often translate this to a 1-9 score in the UC Davis Companion Avian and Exotic Animal Service here. Birds in good physical condition and fully flighted will have robust pectoral musculature, which can correspond to an "overweight" status on a BCS scale. For this reason, some clinicians give both a "keel score" and an overall BCS.



Diet Options for Companion Birds:

Seed-Based Diets

There are various types of diets available to feed birds. For the purposes of this lecture, we will focus on the diets that are most commonly offered to companion birds, such as psittacines, passerine birds, and Columbiformes. The most common commercially available diets can be grouped into four categories: seed-based mixes, formulated diets, pulse mixes, and fresh produce. However, just because we will discuss all of these diets today does NOT mean that they are all created equal or universally recommended.

A commonly encountered diet is comprised of traditional seed-based mixes. Most contain a blend of seeds, nuts, and dried fruit. The types of seeds provided for captive birds contain different kinds of seeds than those found in their native habitats, so they do not well-mimic those encountered in a wild environment. These diets are usually sunflower or safflower seed-based, especially for larger birds, and may or may not be balanced with a pellet supplement. Even when given with pellets, this assumes that the bird is actually consuming the pellet portion of the diet, which many do not, given the higher palatability of the fatty seeds.

Commercial seeds also contain different amino acid profiles and, when husked before eating, lose both fiber and calcium content. Yet they continue to be recommended by the pet trade and are the most readily available avian diets. While birds can survive on these diets, they often develop chronic malnutrition and may show signs of disease once they become more geriatric.

Another consideration is that seed quality is often poor and is not considered fit for human consumption. If not stored correctly, many of these mixes are prone to fungal contamination and growth (especially those containing peanuts), leading to mycotoxin proliferation and exposure, especially when stored in bins or purchased in bulk from a feed store. Most small, packaged containers do not have this problem if fed and consumed promptly.

In general, seed mixes are NOT recommended other than as a portion of the diet in primarily granivorous species. Other food items that are more nutritious and just as enjoyable can be offered. All-seed diets have various qualities that make them suboptimal and the source of many cases of chronic malnutrition:

1. Seeds, especially sunflower-based mixes, contain a high-fat content of >20%. This far exceeds the 5% lipid requirement of most birds. This dramatic increase in fat can predispose birds to obesity, atherosclerosis, and hepatic lipidosis.
2. Inappropriate Ca: Phos ratio and deficient minerals such as iron, copper, manganese, and iodine can lead to hypocalcemia, mineral deficiencies, and nutritional secondary hyperparathyroidism.
3. Low levels of vitamins A, C, D_3, and B can also result in health issues. Hypovitaminosis A is the most commonly encountered clinical issue.
4. These diets are also low in sulfur-containing amino acids (methionine, lysine, cysteine), essential for feather development and growth. Chronic deficiencies result in feather color abnormalities, easily damaged feathers, and dry or hyperkeratotic skin, which can all, in turn, initiate feather-destructive behavior.

Formulated Diets

Formulated diets are made from various foods, including grains, seeds, vegetables, and fruits, and are fortified with vitamins and minerals. Colors, smells, and flavors can be added to increase interest. Some leading brands include Zupreem, Harrison's, Roudybush, Kaytee, and Mazuri, but others also exist. Pellets contain all the needed nutrients in balanced quantities. Selection cannot occur with pellets, and any pellet eaten will have the same nutrient balance. In addition, no additional supplementation of vitamins or minerals should be needed. Many companies have diets formulated for specific species (nectarivores, pigeons, etc). There are also now pellet mixes that have a variety of sizes to enhance enrichment, based on studies of podomandibulation (use of the feet to eat) in larger parrot species. It is important to remember that colorants may be added to improve interest. However, these can also change the color of a bird's feces. This can concern some owners, so it is essential to warn them that this may be a normal change and not indicative of disease! Unique formulations that target specific disease states, such as kidney or liver disease are also available. These may have altered proportions of nutrients, such as decreased protein in renal diets. However, NONE of these prescription diets have been scientifically studied, so keep this in mind when recommending these diets. Disadvantages of pelleted diets to acknowledge include cost, trouble transitioning many birds from seeds to pellets when chronically on suboptimal diets beforehand, and the uniformity that leads to a decreased opportunity for foraging. This could cause boredom in some birds. The uniform appearance of pelleted diets also puts off many owners.

Pulse Mixes

A dietary component used for many South American and frugivorous species is the "pulse mix." Pulses are defined as the dried seed within the pod of legumes. This includes beans such as soybeans, black-eyed beans, and chickpeas. The mix of beans is soaked to allow sprouting and increase digestibility. The sprouted beans are then rinsed to decrease their toxin content before feeding. They can also be a source of bacterial or fungal growth. While many birds enjoy these diets, these mixes are high in carbohydrates and calcium, so additional vitamin and mineral supplementation is required. Likewise, for seed-based diets, the beans used are often not native to the environments from which these birds come.

Fresh Produce

Fresh produce should be a component of most captive avian diets. Non-starchy vegetables should be provided in a higher percentage than fruits or other starchy vegetables. These include dark leafy greens (endive, escarole, mustard greens, dandelion leaves), high in calcium, and herbs (parsley, cilantro), which can be used for enrichment. High-starch vegetables such as potatoes and yams should be avoided. Brassicaceae (kale, broccoli, collard greens, bok choy) can be high in goitrogens (interfere with iodine uptake in the thyroid gland) and should be cooked or only provided in small quantities. Orange and red fruits and vegetables are rich in carotenoids (ß-carotene) and vitamin A and are good additions to captive diets. These include bell pepper, carrots, pumpkin, mango, papaya, and yellow squash. These are especially important in species that rely on these pigments in their feathers for breeding and display behaviors. While most fruits are safe, avocados should NEVER be fed. Avocadoes contain a toxin that causes myocardial degeneration with subsequent heart failure and death, usually within 9-24 hours. This can occur even with ingestion of small amounts.

Supplements

Owners often provide supplements, which can be purchased over the counter. However, excessive supplementation should be avoided when the bird consumes formulated diets. Supplementation in water is an unreliable intake method, as vitamins can degenerate or inactivate in <12hrs, and many birds play with their water. They can also change the taste of the water, causing a bird to avoid drinking and potentially contributing to dehydration. If necessary, supplementation should be provided on food. However, providing additional supplementation is especially risky for fat-soluble vitamins that quickly accumulate to cause toxicity, such as vitamins A and D. There are certain instances when supplementation is recommended:

- Birds weaning from seeds may need additional supplementation during their transition.
- Oral calcium supplementation is recommended for African grey parrots and actively laying birds due to the higher demand on these individuals.
- Iodine supplements such as iodine blocks are only necessary for budgies on a seed diet – pelleted diets contain appropriate amounts of iodine.

Psittacine-Specific Recommendations

Psittacine birds do not need grit, as they husk seeds prior to ingestion. Grit may be a source of minerals, but the negative aspects outweigh the positive – Grit can be over-ingested & cause impactions. It is best to rely on a pelleted diet to provide minerals. Current recommendations for psittacines to provide optimal maintenance = mixture of commercially prepared pellets + fresh produce. In general, 75% of the diet should consist of pellets and 15-25% of the diet should be composed of fresh produce (vegetables > fruit) to create a nutritionally balanced diet. SMALL quantities of HEALTHY human food can be provided for social enrichment but should be less than 5% of the diet overall. Nuts and seeds can be provided, especially in large parrot species, but only 1-2 nuts per day, depending on the size of the bird and the nut. Shelled nuts may provide more significant enrichment and increase fiber content. Examples of acceptable nuts include raw walnuts, Brazil nuts, and almonds, which also act as sources of essential fatty acids. Again, certain nuts, like peanuts, have a higher risk of fungal spores such as Aspergillus sp., so they should be avoided. Seeds can be offered but in VERY limited quantities due to the previously discussed nutritional imbalances and the tendency to select these higher-fat items. When seeds ARE chosen, avoid sunflower seeds as they are higher in fat and have less of the essential fatty acids that are more beneficial. The EXCEPTION is the granivorous species (budgies and passerines) that require higher seed content in their diet. But even in these species, seeds should only account for 25-50% of the diet.

In addition to the TYPE of diet provided, presentation is also important. It is highly recommended that captive birds be provided with foraging opportunities during the day, especially when they are housed alone or spend long periods without an owner. Foraging is an integral part of daily life for wild birds (4-6 hours/day) and can add enrichment and increase energy expenditure in captive birds. Foraging food opportunities include oversized pellets, breakable or pull-apart food items within the cage, and puzzle toys to hide food items.

Diet Conversion:

Converting a captive bird to a pelleted diet can be very challenging, especially in psittacines, due to resistance from the bird and the owner. Older birds on poor diets have become very set in their ways and are used to many tasty, high-fat items. Many owners do not have or want to spend the time to properly convert their birds or feel a special bond associated with feeding certain high-demand foods. Various methods can be proposed to stimulate a diet conversion. However, the most crucial rule is NOT to switch diets suddenly, as birds WILL starve themselves. If this needs to be done, it should be in a hospital setting, where close weight monitoring and nutritional support via gavage feeding can be provided.

Before starting a conversion, it is helpful to monitor the total food intake of the bird to determine the average daily intake. The most common method is to replace 25% of the daily diet with the new diet (pellets + produce) for 1-2 weeks. After those 1-2 weeks, if the bird tolerates the transition well, the latest diet can be increased to 50% of the total diet, and then 75% 1-2 weeks later. The goal is to switch to a 75-100% new diet by 4-6 weeks with only a few seeds/unhealthy dietary items remaining. Numerous pelleted diets are available, so it is important to support owners going through transitions with occasionally stubborn birds and remind them there are different sizes, shapes, colors, and flavors. It is VERY IMPORTANT to monitor a bird's weight and fecal output during this transition to avoid dramatic, rapid weight loss and subsequent illness.

Periods of Increased Nutritional Demand:

Molting

Various situations may naturally require increased nutritional demands throughout a bird's life. One such problem is molting, the normal process of losing and regrowing feathers. Healthy parrots will molt once a year. However, instead of losing all their feathers at once, most psittacines have an incomplete molt, with gradual replacement of feathers (usually symmetrically) to retain the ability to fly. Other avian species, such as ducks, will undergo a “catastrophic molt” in which they lose all their primary feathers.

Feather production is nutritionally demanding, increasing protein requirements by 4-8% (or up to 20% in some species and life stages!). Water requirements can almost double during this time as well. Protein is essential for feather development, especially sulfur-containing amino acids such as methionine, cysteine, and lysine. Feather color is also affected by diet because dietary sources of carotenoids create specific pigment colors such as red, orange, and yellow.

If nutritional demands are unmet, molting may cease, or the interval between molts may change. This results in old, dull, easily damaged feathers. Many birds also develop feather-destructive behavior due to over-grooming or discomfort associated with these changes. Structural colors can be affected when nutrients are deficient, resulting in dull or abnormal coloration. Stress bars can result from the disruption of metabolism during feather formation. It is essential to acknowledge that non-dietary factors may also contribute to structural changes in feather development. A significant point to discuss with owners whose birds are recovering from suboptimal feathering is that, even with dietary improvements, it may take 12-18 months to see outward improvements in the feathers, as they will not be apparent until the successive molt.

Reproduction

Female birds have increased demands for egg laying, and males may also have additional requirements for mating displays and competition. Energy costs for female reproduction can be more than 180% of basal energy requirements in some waterfowl/poultry species. Production species will have higher overall requirements, as they consistently lay eggs. Many companion birds become chronic egg layers and have difficulty maintaining this level of nutritional demand. Females have increased requirements for protein, lipids, calcium, and vitamins. The degree of demand varies by clutch size and frequency of laying, based on the overall market for females. Single-egg clutches do not appear to increase requirements. Generally, improving and adjusting diet about four weeks before egg production is recommended (easier to keep track of in commercial settings such as those used for poultry and waterfowl).

Growth

When young birds hatch, they are considered either precocial or altricial. Precocial birds hatch with eyes open, covered in down, and are ready to leave the nest in 2 days (examples include ducklings and chicks). Altricial young hatch with eyes closed, little or no down, cannot survive outside of the nest, and are fed by their parents (including passerines, parrots, and birds of prey). Altricial young people have increased energy, water, calcium, and protein requirements to support the rapid development of bone and feathers. Energy and protein requirements are highest at hatch (young birds require 20% more protein for growth) and steadily decrease during weaning. Sulfur-containing amino acids (methionine, lysine, cysteine) are in incredibly high demand, as they are needed for feather development. Vitamin D is also essential to support calcium absorption. Water requirements can increase dramatically, with >80% moisture required in the diet. Hand-rearing mixes are often deficient in sulfur-containing amino acids and vitamin D, affecting proper development. Parrot parents will supplement chicks with insects in the wild to increase protein levels.

Calculating Energy Requirements:

Birds, as a whole, expend more energy than mammals. This is due to the demands encountered from previously described life demands plus thermoregulation, flight, and singing. The resting metabolic rate (or RER) in wild birds depends on the climate of the native country of each species, with temperate species having a 20% higher metabolic rate than tropical species. This is not necessarily true in a captive setting. So, how do we actually figure out a bird’s metabolic requirements?

We start with the resting energy requirements (or RER) = **(RER) = k x BW $^{0.75}$**

K = energy constant, specific for that group of animals. Generally, we use an energy constant of 129 for passerine birds and an energy constant of 78 for all other non-passerine species. For reference, mammals typically have an energy constant of 70. Please note this is a highly simplified equation. There are thousands of bird species and various body sizes within those species, so ideally, we would have a K value for each one. While some resources have started compiling this, we clinically use these two main factors as a starting point. From the RER, we can calculate the MER.

MER = maintenance energy requirement = energy *factor* x RER (energy *factor* = specific for an individual's needs). Maintenance is for an adult healthy animal. Factors that will increase overall energy requirements are those mentioned earlier: temperature, growth, reproduction, and molting. Once resting energy requirements are calculated, an energy factor determines the metabolic energy requirement (MER). This energy factor will vary based on the health and life stage of the bird (reflecting current nutritional demands). Some of our more commonly-used energy factors for avian species include:

Energy factors		
Juvenile		1.78
starvation		0.2-0.7
Hypometabolism		0.5-0.9
Elective surgery		1.0-1.2
Mid trauma		1.0-1.2
Severe trauma		1.1-2.0
Growth		1.5-3.0
Sepsis		1.2-1.5
Burns		1.2-2.0
Flight	Aerial (swift)	2.5 - 5.7
	Subtained	11
	Gliding	2
Perching	Resting	1
	Alert	1.9-2.1
Preening		1.6-2.3
Eating		1.7-2.2
Singing, calling		2.9
Grooming, bathing		2.9

Determining basal and daily energy requirements is necessary for making daily dietary recommendations (especially during weight loss plans) and selecting a nutrition/feeding plan for hospitalized and anorectic patients. Calculating MER provides us with the kilocalories a bird requires daily. This can then choose the amount of food to give our bird. Various diets available to support hospitalized patients will depend on their degree of gastrointestinal compromise, age, and feeding strategy. Typical formulations include Lafeber's Emeraid Carnivore care, Omnivore care, or Harrison's Recovery Formula. When feeding thin, malnourished animals, we may want to choose a food with the highest kcal/gram to provide denser nutrition. For this reason, Lafeber Omnivore care is the diet most utilized for our clinical patients, with available kcal being calculated based on the gram or mL amount of diet to be offered. This diet contains 4.09 kcal/g or 1.86 kcal/mL. Specific diet information can often be found on the back of the product container or online by searching for a product's nutritional analysis.

Once we determine the type of diet we would like to feed, we can calculate the volume and frequency based on known nutritional information for these products (often on the back of the container). Dietary requirements can be calculated in grams of diet or mLs of diet. When calculated in grams, we can reconstitute this amount to whatever volume we want, but it is essential to remember our patients' maximum crop volumes (30-50 mL/kg). This is the more accurate calculation method, as water can add another error factor. Many avian patients require nutritional support to maintain weight and health or improve a debilitated status. Hospitalized birds should be gavage fed if they have lost >5% of their body weight in 24 hours or if they continue to lose weight despite eating on their own. Typically, birds can tolerate a crop volume of 30-50 ml/kg. It is essential not to exceed this, as regurgitation and aspiration may occur in a debilitated patient. It is recommended to start at a lower crop volume first (30 ml/kg) and increase as indicated. If a bird tolerates lower volumes well but is still losing weight, this suggests increasing the volume of the offered assist feedings.

When gavage-feeding, it is also essential to determine the frequency based on patient status (can they handle the stress of gavage feeding multiple times per day?) and remember to consider the volume of any medications. Sometimes, it is easier to increase feeding frequency, rather than the volume of each feeding. While some birds are trained to take food or medicines from a syringe or spoon, gavage feeding is the only way to ensure all calories are administered. In psittacines, gavage feeding is provided into the crop through a metal feeding tube. This prevents the bird from biting through the rubber tube, which could lead to a foreign body. In birds of prey and waterfowl, gavage feeding is provided with a red rubber tube into the crop (for example, in hawks and ducks) or directly into the proventriculus (for Strigiformes since they do not have a crop). Given the soft commissure of their beaks, you can use a rubber feeding tube for these species.

IMPORTANT POINTS REGARDING ASSIST-FEEDING:

- ALWAYS stretch the neck and palpate to ensure your tube is in the crop, not the trachea (palpate for two tubes before administering ANY liquid diets or medications).
- Always watch the back of the mouth while administering feedings to monitor for regurgitation.
- If regurgitation occurs, it is best to stop feeding, remove the tube, and place the bird down to allow them to clear the food material themselves.
- Always leave a metal gavage tube attached to the feeding syringe or, if using a red rubber tube, kink the red rubber before removing the feeding tube so liquid does not trickle out and lead to potential aspiration during tube removal.

Nutritional Diseases:

Obesity

Obesity is a significant concern in captive birds due to diets high in fat and a lack of opportunities to exercise. The energy requirements of captive avian species are significantly reduced compared to their wild counterparts. Wild birds spend 50-90% of their time foraging, feeding, and flying, while captive birds do not. This, in addition to selective feeding and inappropriate diets, leads to fat deposition in the body. Some psittacine species are more prone to develop obesity, such as Australian species (Budgerigars, cockatiels, cockatoos) or South American species (Amazons). Obesity is associated with increased low-density cholesterol lipoproteins (LDL). Fat deposition in subcutaneous tissues can lead to lipoma or xanthoma formation, and deposition in coelomic organs can lead to problems such as hepatic lipidosis. Delayed molt and pododermatitis from increased pressure on feet can develop, and these birds may also be predisposed to atherosclerosis. Fat also interacts with calcium and iron and forms insoluble complexes that decrease these minerals' absorption. Prevention of obesity is key. Methods to prevent obesity include gradual caloric restriction for already-overweight birds, providing a nutritionally balanced diet for all, and allowing increased activity and foraging. Keeping birds flighted will help improve exercise and contribute to increased overall welfare.

Hepatic Lipidosis

Hepatic lipidosis, or fatty deposition in the liver and subsequent hepatic dysfunction, is expected in the same species predisposed to obesity. A diet high in saturated fats is the main predisposing factor, but a deficiency of sulfur-containing amino acids (methionine, cysteine, lysine) is also associated with hepatic lipidosis. Birds that become obese can secondarily develop hepatic lipidosis, mainly when lipid is mobilized during periods of hypometabolism or anorexia. Any obese bird exhibiting prolonged anorexia should create concern for the possible development of this condition.

Diagnosis can be challenging. While bloodwork can suggest hepatic disease or dysfunction, abnormalities are not always present and are not a definitive diagnostic tool. Radiographs and coelomic palpation are less sensitive but can indicate hepatomegaly, and ultrasound can be utilized to evaluate for hyperechoic liver parenchyma, suggesting lipid deposition. An ultrasound-guided aspirate is recommended to diagnose hepatic lipidosis. An abundance of lipid droplets is seen in cytology. The most sensitive test to diagnose hepatic lipidosis is histology of a liver biopsy taken during coelomic endoscopy.

Treatment of hepatic lipidosis is symptomatic and based on etiology. The protocol is very similar to feline hepatic lipidosis treatment. Assist feeding, thermal support, and rehydration should be provided as needed. Hepatoprotectants, such as milk thistle and SAMe (denamarin), can be provided to help prevent further damage. Diet changes to reduce lipid and saturated fat intake can be recommended. The prognosis will vary depending on the underlying cause and progression of the disease.

Atherosclerosis

Atherosclerosis describes a thickening of the arterial walls secondary to lipid deposition. It is associated with loss of arterial elasticity, narrowing of the lumen, and rarely thromboembolism. Suggested risk factors in birds include :

- High-energy, high-fat (incredibly saturated) diet such as seeds.
- Dyslipidemia corresponds to an abnormal amount of lipids in the blood, with a higher level of low-density lipid (LDL) and cholesterol or a lower level of high-density cholesterol (HDL)
- Some species, such as African grey parrots, cockatiels, and Amazons, are more prone to developing atherosclerosis. It is also common in galliformes, columbiformes, and some birds of prey (especially in captivity).
- Females over 30 years old have been shown to have a higher risk of atherosclerosis
- Individuals with reduced physical activities are also at increased risk

Clinical signs are associated with cardiac dysfunction/failure and hypertension, including sudden death, congestive heart failure, dyspnea, neurologic signs, respiratory signs, exercise intolerance, and ataxia. Due to the propensity to large vessels, atherosclerosis is often associated with flow-limiting stenosis of major or carotid arteries.

Hypocalcemia and Nutritional Secondary Hyperparathyroidism

Calcium tends to be a primary concern for birds, especially females, due to its high requirements for shell formation. Abnormalities are commonly seen when birds are fed all-seed diets, develop chronic egg laying from environmental stimulation, and/or are kept only indoors with a lack of UVB light.

Calcium deficiency can lead to clinical signs such as:

1. Reproductive abnormalities, including egg binding, shell abnormalities, and cloacal prolapse. Calcium is required for appropriate shell formation and smooth muscle activity during laying. Diets not providing enough calcium may predispose birds to shell abnormalities and egg binding.
2. Seizures, tetany, ataxia, cardiac arrhythmias, and generalized weakness or lethargy. This presentation is most common in very young birds or African grey parrots.
3. Impaired bone development, especially in young, actively growing birds.

African gray parrots are particularly susceptible to hypocalcemia. Total + ionized calcium is commonly run in any African grey parrot presenting for wellness or illness. This species seems to have a higher demand for calcium in their diet, possibly due to variations in protein binding. Diagnosis is made through a blood sample with low ionization and total Ca. Treatment can be in the form of oral or parenteral calcium supplementation. The oral form is the safest, and caution should be used with prolonged parenteral supplementation due to the risk of hypercalcemia. Parenteral calcium can be used initially when ionized calcium is deficient, and then the bird can be transitioned to oral calcium. Dietary supplementation can also be provided (such as cuttlebones), and some food companies offer diets that contain higher calcium for susceptible species like African grey parrots.

Vitamin D is critical in calcium metabolism, bone mineralization, and reproduction (eggs) and can be stored in the uropygial (preen) gland. Vitamin D_3 is a bioactive form that can be absorbed through diet or activated through UVB light. Thus, UVB light may help in the subsequent absorption of calcium. Suggested sufficient exposure is only 30 min/day for most captive birds. However, it is not a requirement as no studies have shown a definitive benefit. As a reminder, calcium metabolism is regulated by Vitamin D, parathyroid hormone, and calcitonin. Included here is a little chart summarizing the functions of the vitamins regarding Ca and P homeostasis:

Hormone	**Ca**	**P**
Vitamin D	↑	↑
Parathyroid Hormone (PTH)	↑	↓
Calcitonin	↓	↓

Calcium and vitamin D deficiency can result in nutritional secondary hyperparathyroidism (or NSHP). Manifestations of this in birds include juvenile osteodystrophy, seizures, fractures, and poor reproductive performance. Osteodystrophy in chicks presents with limb bowing, pathologic fractures, and other skeletal deformities. Surgical intervention may be warranted, but it is difficult because of the bones' weakness.

Hypovitaminosis A

Vitamin A is vital for vision, reproduction, immunity, cellular differentiation, growth, and embryonic development. Carotenoid precursors commonly found in plants are converted to vitamin A by intestinal enzymes. Vitamin A is then conserved in the liver once ingested. Vitamin A deficiency is one of the most common early problems detected in birds on all-seed diets. Vitamin A deficiency causes squamous metaplasia and hyperkeratosis of epithelial cells. The main clinical signs on physical examination are blunted choanal papilla, plaque or abscess in the oral cavity, poor feather quality, pododermatitis, or uropygial gland impaction to abscessation. In very advanced cases, renal failure can also be a consequence of hypovitaminosis A, as well as digestive malabsorption, air sac hyperinflation by accumulation of cell debris, and predisposition to recurrent upper respiratory infections.

Improving the diet's carotenoid and vitamin A content by providing more dark-leafy greens and orange-colored produce can enhance the amount of vitamin A the bird synthesizes endogenously. While oral and parenteral supplementation can be provided, caution should be taken with parenteral vitamin A supplementation due to the risk of hypervitaminosis A.

Vitamin A toxicity can be commonly encountered when supplementation is provided and can occur with moderate and high dietary supplementation levels. So, there is typically an iatrogenic etiology to toxicity, rather than the bird consuming too many pellets, for example. Clinical signs of hypervitaminosis A can be similar to hypovitaminosis A and can also include altered vocalization patterns, poor feather quality, loss of muscle mass, behavioral changes, poor reproductive performance, and recurrent infections due to loss of epithelial integrity.

Iodine Deficiency (Goiter)

Chronic iodine deficiency can be seen in budgerigars fed all-seed diets not supplemented with additional iodine. Current pelleted and formulated seed diets should contain appropriate amounts of iodine, so this is not as common anymore. It is also not commonly seen in larger parrot species. Iodine deficiency leads to thyroid hyperplasia and enlargement of the thyroid gland, known as goiter. This leads to a visible swelling in the clavicular region and subsequent voice changes, regurgitation, and dyspnea due to intrathoracic compression of the esophagus and syrinx. Obesity can also be seen due to changes in metabolic rate.

Diagnosis is usually presumptive based on history, physical examination, and response to treatment. Even if markedly enlarged, the thyroid gland in birds lies within their coelomic cavity, so it is not ordinarily palpable or noted on physical examination. Treatment requires iodine supplementation, which can be done in the water (Lugol’s iodine) or via pink iodine blocks. Supplementation in water requires the bird to drink enough water, so it is essential to keep this in mind when formulating one’s treatment plan.